Real-Time Human Activity Recognition Using Time-Series Transformers

Overview

This project implements real-time Human Activity Recognition (HAR) using Transformer-based deep learning models trained on wearable sensor data from the HAR-70 dataset.

Features

- Transformer model for multivariate time-series classification

- Real-time streaming activity recognition

- Noise filtering and feature engineering

- Modular ML pipeline for training and inference

Dataset

The HAR-70 dataset includes:

- 70 subjects

- IMU sensor data: accelerometer, gyroscope, (optional magnetometer)

- Windowed sequences for deep learning input

Model Architecture

The model uses:

- Positional encoding

- Multi-head attention

- Feed-forward networks

- Softmax classification head

Training

Training is performed using Python scripts that:

- Preprocess data

- Train transformer models

- Save checkpoints

Real-Time Demo

A script allows real-time inference through streaming sensor or synthetic data.

Installation

1. Clone the repository

2. Install dependencies using requirements.txt

3. Run training or inference scripts

Future Work

- On-device inference (TFLite)

- Attention visualization

- Hyperparameter optimization